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Subject: DBMS

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Experiment No. 3

Aim: Create and populate database using Data Definition Language (DDL) Apply Integrity Constraints for the specified system.

Output:

1.CREATE

```
mysql> use food_deliveryy;
Database changed
mysql> CREATE TABLE Restaurant (
->   restaurant_id INT PRIMARY KEY,
->   name VARCHAR(100) NOT NULL,
->   location VARCHAR(100) NOT NULL,
->   rating DECIMAL(2,1) CHECK (rating >= 0 AND rating <= 5)
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE TABLE Customer (
->   cust_id INT PRIMARY KEY,
->   cust_name VARCHAR(100) NOT NULL,
->   email VARCHAR(100) UNIQUE,
->   phone_no VARCHAR(15) UNIQUE,
->   address VARCHAR(200)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Orders (
->   order_id INT PRIMARY KEY,
->   cust_id INT,
->   restaurant_id INT,
->   order_date DATETIME DEFAULT CURRENT_TIMESTAMP,
->   total_amount DECIMAL(10,2) CHECK (total_amount > 0),
->   FOREIGN KEY (cust_id) REFERENCES Customer(cust_id),
->   FOREIGN KEY (restaurant_id) REFERENCES Restaurant(restaurant_id)
-> );
Query OK, 0 rows affected (0.06 sec)
```



2.Alter command:

a) Adding column to table

```
mysql> ALTER TABLE Customer
  -> ADD city VARCHAR(50);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Customer;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| cust_id | int | NO | PRI | NULL |  |
| cust_name | varchar(100) | NO |  | NULL |  |
| email | varchar(100) | YES | UNI | NULL |  |
| phone_no | varchar(15) | YES | UNI | NULL |  |
| address | varchar(200) | YES |  | NULL |  |
| city | varchar(50) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.05 sec)
```

b) Dropping the column:

```
mysql> ALTER TABLE Customer
  -> DROP city;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Customer;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| cust_id | int | NO | PRI | NULL |  |
| cust_name | varchar(100) | NO |  | NULL |  |
| email | varchar(100) | YES | UNI | NULL |  |
| phone_no | varchar(15) | YES | UNI | NULL |  |
| address | varchar(200) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

c) Rename the column name:

```
mysql> ALTER TABLE Customer
  -> CHANGE cust_name customer_name VARCHAR(100);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Customer;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| cust_id | int | NO | PRI | NULL |  |
| customer_name | varchar(100) | YES |  | NULL |  |
| email | varchar(100) | YES | UNI | NULL |  |
| phone_no | varchar(15) | YES | UNI | NULL |  |
| address | varchar(200) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```



d) Modify the datatype of column:

```
mysql> ALTER TABLE Customer
-> MODIFY phone_no VARCHAR(20);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Customer;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| cust_id    | int           | NO   | PRI | NULL    |       |
| customer_name | varchar(100)  | YES  |     | NULL    |       |
| email      | varchar(100)  | YES  |     | NULL    |       |
| phone_no   | varchar(20)   | YES  |     | NULL    |       |
| address    | varchar(200)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

3) Rename command: -

```
mysql> RENAME TABLE Customer TO Users;
Query OK, 0 rows affected (0.03 sec)

mysql> desc Customer;
ERROR 1146 (42S02): Table 'food_deliveryy.customer' doesn't exist
mysql> desc Users;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| cust_id    | int           | NO   | PRI | NULL    |       |
| customer_name | varchar(100)  | YES  |     | NULL    |       |
| email      | varchar(100)  | YES  |     | NULL    |       |
| phone_no   | varchar(20)   | YES  |     | NULL    |       |
| address    | varchar(200)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

4) Drop command: -

```
mysql> DROP TABLE Users;
ERROR 1146 (42S02): Cannot drop table 'Users' referenced by a foreign key constraint 'orders_ibfk_1' on table 'orders'.
mysql> TRUNCATE TABLE Users;
ERROR 1146 (42S02): Cannot truncate a table referenced in a foreign key constraint ('food_deliveryy'.'orders', CONSTRAINT 'orders_ibfk_1')
mysql> DROP TABLE Orders;
Query OK, 0 rows affected (0.03 sec)

mysql> DROP TABLE Users;
Query OK, 0 rows affected (0.03 sec)

mysql> show tables;
+-----+
| Tables_in_food_deliveryy |
+-----+
| restaurant              |
+-----+
1 row in set (0.03 sec)
```

5) Truncate command: -

```
mysql> TRUNCATE TABLE Orders;
ERROR 1146 (42S02): Table 'food_deliveryy.orders' doesn't exist
mysql> CREATE TABLE Users (
-> cust_id INT PRIMARY KEY,
-> customer_name VARCHAR(100)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> TRUNCATE TABLE Users;
Query OK, 0 rows affected (0.04 sec)

mysql> |
```



1. NOT NULL

```
mysql> CREATE TABLE Customer (  
->   cust_id INT PRIMARY KEY,  
->   cust_name VARCHAR(50) NOT NULL  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO Customer VALUES (1, 'Rahul');  
Query OK, 1 row affected (0.03 sec)  
  
mysql> INSERT INTO Customer VALUES (2, NULL);  
ERROR 1048 (23000): Column 'cust_name' cannot be null  
mysql>
```

DOMAIN Constraints:

```
mysql> INSERT INTO Users VALUES (101, 25);  
Query OK, 1 row affected (0.04 sec)
```

```
mysql> INSERT INTO Users VALUES ('One', 'Twenty');  
ERROR 1366 (HY000): Incorrect integer value: 'One' for column 'cust_id' at row 1  
mysql>
```

PRIMARY KEY Constraints:

```
mysql> INSERT INTO restaurant VALUES(3, 'FoodClub', 'THANE', 3);  
ERROR 1062 (23000): Duplicate entry '3' for key 'restaurant.PRIMARY'  
mysql>
```

FOREIGN KEY Constraint:

```
mysql> INSERT INTO Orders VALUES (102, 5);  
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('food_deliveryy'. 'orders', CONSTRA  
INT 'orders_ibfk_1' FOREIGN KEY ('cust_id') REFERENCES 'customer' ('cust_id'))  
mysql>
```

CHECK Constraint

```
mysql> CREATE TABLE Payment (  
->   payment_id INT PRIMARY KEY,  
->   amount DECIMAL(8,2),  
->   CHECK (amount > 0)  
-> );  
Query OK, 0 rows affected (0.06 sec)  
  
mysql> INSERT INTO Payment VALUES (1, 500);  
Query OK, 1 row affected (0.02 sec)  
  
mysql> INSERT INTO Payment VALUES (2, -100);  
ERROR 3819 (HY000): Check constraint 'payment_chk_1' is violated.  
mysql>
```



Exercise 1:

```
Command Prompt - mysql -u X + v

Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASHMI>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.43 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use database food_deliver;
ERROR 1049 (42000): Unknown database 'database'
mysql> use food_delivery;
Database changed
mysql> CREATE TABLE Customer (
  ->   cust_id INT PRIMARY KEY,
  ->   first_name VARCHAR(50) NOT NULL,
  ->   last_name VARCHAR(50) NOT NULL,
  ->   address VARCHAR(200) NOT NULL,
  ->   phone_no VARCHAR(15) UNIQUE NOT NULL,
  ->   email VARCHAR(100) UNIQUE
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE Restaurant (
  ->   restaurant_id INT PRIMARY KEY,
  ->   name VARCHAR(100) NOT NULL,
  ->   location VARCHAR(100) NOT NULL,
  ->   rating DECIMAL(2,1) CHECK (rating BETWEEN 0 AND 5)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Menu_Items (
  ->   item_id INT PRIMARY KEY,
  ->   restaurant_id INT,
  ->   name VARCHAR(100) NOT NULL,
  ->   description VARCHAR(200),
  ->   price DECIMAL(6,2) CHECK (price > 0),
  ->   FOREIGN KEY (restaurant_id)
  ->   REFERENCES Restaurant(restaurant_id)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE Delivery_Agent (
  ->   agent_id INT PRIMARY KEY,
  ->   first_name VARCHAR(50) NOT NULL,
  ->   last_name VARCHAR(50) NOT NULL,
  ->   phone_no VARCHAR(15) UNIQUE NOT NULL,
  ->   vehicle_no VARCHAR(20) UNIQUE,
  ->   salary DECIMAL(8,2) CHECK (salary > 0)
  -> );
Query OK, 0 rows affected (0.04 sec)
```




```
mysql> CREATE TABLE Orders (  
->   order_id INT PRIMARY KEY,  
->   cust_id INT,  
->   restaurant_id INT,  
->   agent_id INT,  
->   time_of_order TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
->   status VARCHAR(20) DEFAULT 'Pending',  
->  
->   FOREIGN KEY (cust_id)  
->   REFERENCES Customer(cust_id),  
->  
->   FOREIGN KEY (restaurant_id)  
->   REFERENCES Restaurant(restaurant_id),  
->  
->   FOREIGN KEY (agent_id)  
->   REFERENCES Delivery_Agent(agent_id)  
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> CREATE TABLE Items (  
->   item_id INT PRIMARY KEY,  
->   name VARCHAR(100) NOT NULL,  
->   category VARCHAR(50),  
->   price DECIMAL(6,2) CHECK (price > 0),  
->   quantity INT DEFAULT 1 CHECK (quantity >= 0)  
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> CREATE TABLE Payment (  
->   payment_id INT PRIMARY KEY,  
->   order_id INT UNIQUE,  
->   type VARCHAR(20) NOT NULL,  
->   status VARCHAR(20) DEFAULT 'Pending',  
->   time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
->  
->   FOREIGN KEY (order_id)  
->   REFERENCES Orders(order_id)  
-> );
```

Query OK, 0 rows affected (0.14 sec)

```
mysql> desc Customer;
```

Field	Type	Null	Key	Default	Extra
cust_id	int	NO	PRI	NULL	
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
address	varchar(200)	NO		NULL	
phone_no	varchar(15)	NO	UNI	NULL	
email	varchar(100)	YES	UNI	NULL	

6 rows in set (0.05 sec)

```
mysql> desc Restaurant;
```

Field	Type	Null	Key	Default	Extra
restaurant_id	int	NO	PRI	NULL	
name	varchar(100)	NO		NULL	
location	varchar(100)	NO		NULL	
rating	decimal(2,1)	YES		NULL	



```
mysql> desc Menu_Items;
```

Field	Type	Null	Key	Default	Extra
item_id	int	NO	PRI	NULL	
restaurant_id	int	YES	MUL	NULL	
name	varchar(100)	NO		NULL	
description	varchar(200)	YES		NULL	
price	decimal(6,2)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> desc Delivery_Agent;
```

Field	Type	Null	Key	Default	Extra
agent_id	int	NO	PRI	NULL	
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
phone_no	varchar(15)	NO	UNI	NULL	
vehicle_no	varchar(20)	YES	UNI	NULL	
salary	decimal(8,2)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> desc Orders;
```

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	
cust_id	int	YES	MUL	NULL	
restaurant_id	int	YES	MUL	NULL	
agent_id	int	YES	MUL	NULL	
time_of_order	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
status	varchar(20)	YES		Pending	

```
6 rows in set (0.05 sec)
```

```
mysql> desc Items;
```

Field	Type	Null	Key	Default	Extra
item_id	int	NO	PRI	NULL	
name	varchar(100)	NO		NULL	
category	varchar(50)	YES		NULL	
price	decimal(6,2)	YES		NULL	
quantity	int	YES		1	

```
5 rows in set (0.05 sec)
```

```
mysql> desc Payment;
```

Field	Type	Null	Key	Default	Extra
payment_id	int	NO	PRI	NULL	
order_id	int	YES	UNI	NULL	
type	varchar(20)	NO		NULL	
status	varchar(20)	YES		Pending	
time	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED



Exercise 2:

```
mysql> INSERT INTO Customer VALUES
  -> (1,'Rahul','Sharma','Mumbai','9876543210','rahul@gmail.com'),
  -> (2,'Anita','Patil','Pune','9876543211','anita@gmail.com');
Query OK, 2 rows affected (0.13 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Restaurant VALUES
  -> (101,'Spice Garden','Mumbai',4.5),
  -> (102,'Food Hub','Pune',4.2);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Menu_Items VALUES
  -> (201,101,'Paneer Butter Masala','Indian Dish',250),
  -> (202,102,'Veg Burger','Fast Food',120);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Delivery_Agent VALUES
  -> (301,'Ravi','Kumar','9123456780','MH12AB1234',15000),
  -> (302,'Suresh','Patil','9123456781','MH14XY5678',14000);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Orders VALUES
  -> (401,1,101,301,'2026-03-08 18:30:00','Confirmed'),
  -> (402,2,102,302,'2026-03-08 19:00:00','Pending');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Items VALUES
  -> (501,'Paneer Butter Masala','Main Course',250,2),
  -> (502,'Veg Burger','Fast Food',120,1);
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```




```
mysql> INSERT INTO Payment VALUES
  -> (601,401,'UPI','Completed','2026-03-08 18:35:00'),
  -> (602,402,'Cash','Pending','2026-03-08 19:05:00');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Customer VALUES
  -> (3,NULL,'Verma','Delhi','9876543212','verma@gmail.com');
ERROR 1048 (23000): Column 'first_name' cannot be null
mysql> INSERT INTO Customer VALUES
  -> (4,'Amit','Joshi','Delhi','9876543210','amit@gmail.com');
ERROR 1062 (23000): Duplicate entry '9876543210' for key 'customer.phone_no'
mysql> INSERT INTO Restaurant VALUES
  -> (103,'Taste Point','Delhi',7);
ERROR 3819 (HY000): Check constraint 'restaurant_chk_1' is violated.
mysql> INSERT INTO Orders VALUES
  -> (403,10,101,301,'2026-03-08 20:00:00','Pending');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('food_delivery'. 'orders', CONSTRAINT 'orders_ibfk_1' FOREIGN KEY ('cust_id') REFERENCES 'customer' ('cust_id'))
mysql> SELECT * FROM Customer;
+-----+-----+-----+-----+-----+-----+
| cust_id | first_name | last_name | address | phone_no | email |
+-----+-----+-----+-----+-----+-----+
| 1 | Rahul | Sharma | Mumbai | 9876543210 | rahul@gmail.com |
| 2 | Anita | Patil | Pune | 9876543211 | anita@gmail.com |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql> SELECT * FROM Restaurant;
+-----+-----+-----+-----+
| restaurant_id | name | location | rating |
+-----+-----+-----+-----+
| 101 | Spice Garden | Mumbai | 4.5 |
| 102 | Food Hub | Pune | 4.2 |
+-----+-----+-----+-----+
```



```
mysql> SELECT * FROM Menu_Items;
+-----+-----+-----+-----+-----+
| item_id | restaurant_id | name | description | price |
+-----+-----+-----+-----+-----+
| 201 | 101 | Paneer Butter Masala | Indian Dish | 250.00 |
| 202 | 102 | Veg Burger | Fast Food | 120.00 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Orders;
+-----+-----+-----+-----+-----+-----+
| order_id | cust_id | restaurant_id | agent_id | time_of_order | status |
+-----+-----+-----+-----+-----+-----+
| 401 | 1 | 101 | 301 | 2026-03-08 18:30:00 | Confirmed |
| 402 | 2 | 102 | 302 | 2026-03-08 19:00:00 | Pending |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Payment;
+-----+-----+-----+-----+-----+
| payment_id | order_id | type | status | time |
+-----+-----+-----+-----+-----+
| 601 | 401 | UPI | Completed | 2026-03-08 18:35:00 |
| 602 | 402 | Cash | Pending | 2026-03-08 19:05:00 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Delivery_Agent;
+-----+-----+-----+-----+-----+-----+
| agent_id | first_name | last_name | phone_no | vehicle_no | salary |
+-----+-----+-----+-----+-----+-----+
| 301 | Ravi | Kumar | 9123456780 | MH12AB1234 | 15000.00 |
| 302 | Suresh | Patil | 9123456781 | MH14XY5678 | 14000.00 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Conclusion: This experiment helped in understanding the use of DDL commands to create and manage a relational database. By applying integrity constraints, the database ensured data accuracy and consistency. It demonstrated how proper database design enforces rules and supports efficient data management.